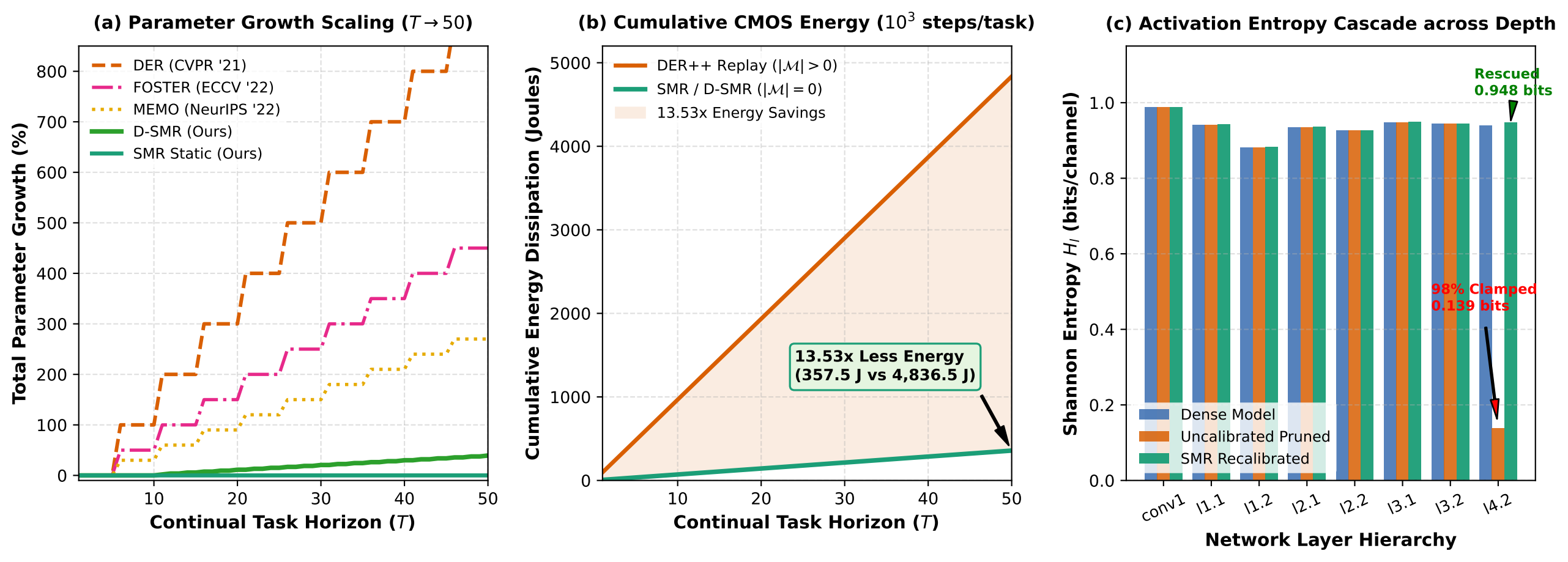
(a) Parameter Growth Scaling ($T \to 50$)
DER (CVPR '21)
FOSTER (ECCV '22)
MEMO (NeurIPS '22)
D-SMR (Ours)
SMR Static (Ours)
Total Parameter Growth (%)
Continual Task Horizon (T)
(b) Cumulative CMOS Energy (10^3 steps/task)
DER++ Replay ($|\mathcal{M}| > 0$)
SMR / D-SMR ($|\mathcal{M}| = 0$)
13.53x Energy Savings
13.53x Less Energy
(357.5 J vs 4,836.5 J)
Cumulative Energy Dissipation (Joules)
Continual Task Horizon (T)
(c) Activation Entropy Cascade across Depth
Rescued
0.948 bits
98% Clamped
0.139 bits
Dense Model
Uncalibrated Pruned
SMR Recalibrated
Shannon Entropy H_l (bits/channel)
conv1
l1.1
l1.2
l2.1
l2.2
l3.1
l3.2
l4.2
Network Layer Hierarchy